Homework 1: Due Oct 19 at 11:59pm Montreal time

General guidelines for homeworks:

- You are encouraged to meet with other students to discuss the homework, but all write-ups must be done on your own. Do not take notes from those meetings. You should know how to work out the solutions by yourself.
- Please acknowledge other students with whom you discussed the problems and what resources (other than the instructor/TAs, lecture notes, and textbook) you used to help you solve the problem. This won't affect your grade.
- Homework grades will be based not only on getting the "correct answer", but also on good writing style and clear presentation of your solution. It is your responsibility to make sure that the marker can easily follow your line of reasoning.
- For programming questions, please make sure your code is clearly commented and easy to read. If the marker cannot understand your code and it does not run correctly, you might not be able to get any partial marks.
- Follow the instructions exactly. We reserve the right to refuse to grade the homework or deduct marks if the instructions are not followed.
- Use your grace days wisely.

1 Probability and Bayes rule (10 points)

A new band called B_2 is inspired by an old band called B_1. 90% of music critics think B_1 was a great (G) band, 10% that it was moderate (M) and 0% that it was aweful (A). These critics have also compiled the following table:

		B_2		
		G	M	A
B_1	G	1	0	0
	M	0.1	0.9	0
	A	0.2	0.3	0.5

The table says that the probability of a new band (B_2) being great given that the inspiring band (B_1) was great is $P(B_2 = G|B_1 = G) = 1$. Similarly, $P(B_2 = G|B_1 = M) = 0.1$, $P(B_2 = M|B_1 = A) = 0.3$, and so on. Note that the numbers in each row sum to 1.

(a) Given what the critics think of B_1 and the fact that B_1 inspired B_2, what is the probability that B_2 is a great band? i.e. $P(B_2 = G) =?$

(b) What is $P(B_1 = G|B_2 = G)$ in this example?

2 MLE and MAP (15 points)

The Poisson distribution is useful for modeling the number of events occurring within a unit time, such as the number of packets arrived at some server per minute. The probability mass function of a Poisson distribution is as follows:

$$P(k|\lambda) = \frac{\lambda^k e^{-\lambda}}{k!}$$

where $\lambda > 0$ is the parameter of the distribution and $k \in \{0, 1, 2, ...\}$ is the discrete random variable modeling the number of events encountered per unit time.

(a) Let $\{k_1, k_2, ..., k_n\}$ be i.i.d samples drawn from a Poisson distribution with parameter λ. Derive the MLE estimate $\hat{\lambda}_{MLE}$.

(b) Suppose you believe the Gamma distribution:

$$P(\lambda) = \frac{\lambda^{\alpha-1} e^{-\lambda/\beta}}{\Gamma(\alpha)\beta^\alpha}$$

is a good prior for λ, where $\Gamma(\cdot)$ is the Gamma function, and you know the values of the two hyper-parameters $\alpha > 1$ and $\beta > 0$. Derive the MAP estimate $\hat{\lambda}_{MAP}$.

(c) What happens to $\hat{\lambda}_{MAP}$ when the sample size n goes to infinity? How does it relate to $\hat{\lambda}_{MLE}$?

3 Ridge regression (5 points)

Show that the ridge regression estimator $\hat{\mathbf{w}}_{ridge} = (\lambda I + X^T X)^{-1} X^T \mathbf{t}$ is the the solution that minimizes the following regularized least square function. (hint: $\mathbf{w}^T \mathbf{w} = \mathbf{w}^T I \mathbf{w}$)

$$E_{ridge}(\mathbf{w}) = (X\mathbf{w} - \mathbf{t})^T (X\mathbf{w} - \mathbf{t}) + \lambda \mathbf{w}^T \mathbf{w}$$

4 Programming: Nonlinear Regression (15 points)

In this question you will implement linear basis function regression with polynomial and Gaussian basis functions. You will need the code and data in the folder "`polynomial`". The dataset is the `AutoMPG` dataset from the UCI repository. The task is to predict fuel efficiency (miles per gallon) from 7 features describing a car.

Functions are provided for loading the data[1], and normalizing the features and targets to have 0 mean and unit variance.

```
[t,X] = loadData();
X_n = normalizeData(X);
t = normalizeData(t);
```

[1]Note that loadData reorders the datapoints using a fixed permutation. Use this fixed permutation for the questions in this assignment. If you are interested in what happens in "reality", try using a random permutation afterwards. Results will not always be as clean as you will get with the fixed permutation provided.

For the following, use these normalized features `X_n` and targets.

Implement linear basis function regression with polynomial basis functions. For a given input $\mathbf{x} = [x_1, x_2, ...]^T$, use only monomials of a single variable in $\mathbf{x}$ (e.g., x_1, x_1^2, x_2^2) and no cross-terms ($x_1 \cdot x_2$). You may also find the provided function `degexpand` in `utils.py` useful for generating the basis functions.

Perform the following experiments:

(a) Create a Python script `polynomial_regression.py` for the following.

Using the first 100 points as training data, and the remainder as testing data, fit a polynomial basis function regression for degree 1 to degree 10 polynomials (note that you can use `degexpand` to create the polynomial terms). Do not use any regularization. Plot training error and test error (i.e. mean squared error) versus polynomial degree. **Put this plot, along with a brief comment on what you see, in your report.**

(b) Create a Python script `polynomial_regression_1d_vis.py` for the following.

It is difficult to visualize the results of high-dimensional regression. Instead, only use one of the features (use the 3rd feature, i.e. `X_n[:,2]` since Python index starts with 0) and again perform polynomial regression. Produce plots of the training data points, learned polynomial, and test data points. The code `visualize_1d.py` may be useful. **Put 2 or 3 of these plots, for interesting (low-order, high-order) results, in your report. Include brief comments.**

(c) Create a Python script `polynomial_regression_reg.py` for the following.

Implement L_2-regularized regression. Again, use the first 100 points, and only use the 3rd feature. Fit a degree 8 polynomial using $\lambda = \{0, 0.01, 0.1, 1, 10, 100, 1000\}$. Use 10-fold cross-validation to decide on the best value for λ. Produce a plot of average validation set error versus regularizer value. Use a `semilogx` plot, putting regularizer value on a log scale[2]. **Put this plot in your report, and note which regularizer value you would choose from the cross-validation.**

How to Submit

Please create a document in PDF format (named it as `hw1.pdf`) containing the following:

- Your name, student ID, email address
- Your solutions to the questions (including plots/figures)

You are strongly encouraged to type the solutions. But you really prefer, you can also write the solutions by hand, then scan them and create a PDF document. If you choose to write your solutions by hand, please make sure it is neat and legible. If the markers cannot understand your handwriting, you will lose marks.

Put your codes (including data files and others that are provided as part of the assignment) in a folder named `lastname_firstname` (replace with your last and first names). Create a ZIP file (name

[2]The unregularized result will not appear on this scale. You can either add it as a separate horizontal line as a baseline, or report this number separately.

it `lastname_firstname.zip`, again, replace with your last and first names accordingly) containing the top-level directory.

For example, if I were to submit the homework, I would create a ZIP file named `wang_yang.zip`. After running "`unzip wang_yang.zip`", I should get the following:

```
wang_yang/
wang_yang/polynomial_regression.py
wang_yang/polynomial_regression_1d_vis.py
wang_yang/...... (other files)
```

If I run `python polynomial_regression.py` in the `wang_yang` folder, I should get the plots in the report from the relevant question.

Go to this course in Moodle, then click "`HW1 submission`". Submit the following two files: 1) the PDF document; 2) the zip file containing your codes (see above). The submission server is configured to only accept PDF and ZIP files. If you try to submit other file format, the submission server will not accept it. So make sure you know how to create PDF and ZIP files ahead of time.

Note that you can make multiple submissions. But we will only consider the last submission. We will use the time stamp recorded on Moodle to calculate late days you have used for this assignment.